Lifestyle Intervention and Urbanization Effects on Obesity in China

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1. Revised PICO Question:

Among adults with overweight or obesity in China, does participating in a structured lifestyle intervention that includes dietary modification and increased physical activity, compared to usual care or no intervention lead to a measurable reduction on obesity related metabolic complications such as metabolic syndrome and diabetes risk factors over a 12 month period.

PICO Revision Explanation:

This revised PICO question was refined from a broader focus on obesity trends to a more specific emphasis on structured lifestyle interventions and measurable metabolic outcomes. The updated version clarifies the comparison group (usual care or no intervention) and specifies a defined time frame of 12 months, improving clarity and evaluability.

Background:

Obesity in China has increased rapidly alongside economic growth and urbanization, making it a major public health concern. Between 2014 and 2018, China had the largest number of adults with obesity worldwide, with over 90 million individuals affected (Mu et al., 2021). Obesity increases the risk of chronic diseases such as type 2 diabetes, hypertension, and cardiovascular disease, leading to long-term health complications and early mortality. Urbanization has reduced transportation-related physical activity, contributing to more sedentary lifestyles, as seen with the rise in automobile use (Chen, 2023).

The rise in obesity also has significant economic impacts, with obesity-related healthcare costs projected to reach 480 million yuan by 2030, representing 21.5% of total national medical expenditures (Ting et al., 2024). Therefore, it is important to evaluate whether structured lifestyle interventions targeting diet and physical activity can reduce metabolic risk factors and inform public health policy.

Design Decision Memo

Objectives and Purpose:

The purpose of this study is to examine how urbanization related lifestyle changes are associated with obesity prevalence among adults in China, with a focus on transportation-related physical activity and structural environmental changes.

SMART Objectives:

1. To assess the association between transportation related physical activity and obesity status among adults using CHNS data within a defined analytical period.
2. To evaluate how community level urbanization indicators, including motor vehicle ownership and urbanization index scores, are associated with obesity prevalence while controlling for demographic variables such as age, sex, and income.

Study Design

This project uses observational research design, specifically a cross sectional analysis of secondary data sources related to obesity prevalence, transportation trends, and urbanization indicators in China. Existing data from national health surveys, population studies, and published literature will be used to examine patterns and associations between urban lifestyle changes and obesity rates.

A cross section design allows the study to analyze relationships between variables at a particular period in time. By examining the previously collected national level data and epidemiological studies. The project can identify patterns linking urbanization related lifestyle shifts with increasing obesity prevalence.

Justification for Selected Design

An observational cross sectional design is appropriate for this research question because the study focuses on population level patterns and trends rather than testing a specific intervention. The goal is to understand how large scale societal changes, such as economic development and urbanization are associated with changes in health outcomes.

Furthermore, this design allows researchers to analyze large population datasets that are available. National surveys and published epidemiological studies provided reliable data on obesity prevalence, physical activity patterns, and demographic changes. This makes the design both practical and efficient for examining the public health implication of urbanization.

Rigorous Experimental Designs Were Not Feasible

More rigorous experimental designs such as randomized controlled trials, were not feasible or appropriate for this research question. Randomized trials are used to test the effects of specific interventions under controlled conditions. However, the exposure of interest is urbanized and transportation related lifestyle changes, which occur at social and environmental level rather than individual level.

Furthermore, it would not be possible to randomly assign individuals or communities to experience different levels of urban development, economic growth, or transportation infrastructure. These factors are determined by broader economic and policy changes that occur naturally over time.

Since the research question focuses on understanding real world population trends rather than testing a controlled intervention, an observational design is more appropriate for examining how these structural changes influence health outcomes.

Key Limitations

Although the observational cross sectional design is appropriate for this study, it has several limitations. First, cross sectional can identify associations between variables but is unable to find clear causal relationships. While the analysis may show that reduced physical activity and increased automobile use are associated with higher obesity rates, the design cannot definitely prove that these factors directly cause the increase.

Secondly, the secondary data sources may vary in how variables are measured across studies or surveys. Differences in measurement methods, sampling strategies, or definitions of obesity could affect the consistency of the data.

Finally, obesity is influenced by many factors beyond physical activity and transportation changes. Dietary shifts, income growth, food availability, and cultural changes may also contribute to rising obesity rates. Because of these multiple influences, findings from this study should be interpreted as identifying important contributing factors rather than definitive explanations.

Despite these limitations, the design provides valuable insights into how large scale social and environmental changes are associated with populations health outcomes. Understanding these relationships is important for informing future public health policies and interventions aimed at preventing obesity and related chronic diseases.

3. Data Collection and Measurement Plan

Data Sources

This study will rely on secondary data sources to examine the relationship between urbanization related lifestyle changes and obesity in China. The primary dataset used will be the China Health and Nutrition Survey (CHNS). CHNS is a longitudinal household survey conducted across multiple Chinese provinces that collects detailed information on physical

activity, transportation behaviors, socioeconomic status, and anthropometric measurements. The dataset is appropriate for this study because it includes measurement of height and weight as well as accurate calculations of body mass index (BMI).

Furthermore, contextual indicators of urbanization will be obtained from the National Bureau of Statistics of China, this includes measurements such as urban population proportion, motor vehicle ownership rates, and urban population proportion, motor vehicle ownership rates, and regional economic development indicators. These variables provide community level measures of urbanization that can be linked to individual level health data from CHNS.

Using secondary data allows the study to analyze large population samples across multiple regions and years, which is necessary for examining national trends in obesity and lifestyle changes. A major strength of using the CHNS dataset is its large, nationally representative sample and use of standardized anthropometric measurements, which improves reliability and generalizability of findings. A key limitation is reliance on self reported physical activity data, which may introduce recall bias and reduce the accuracy of measurements.

Measurement Tools

The primary measurement tool will be the CHNS survey dataset which contains structured questionnaire data and anthropometric measurements that are collected by field staff.

Obesity will be measured using BMI calculated from height and weight. Height is recorded using a portable stadiometer and weight from calibrated digital scales during the survey. BMI will be calculated using the standard formula:

$$\text{BMI} = \text{weight (kg)}/\text{height(m^2)}$$

Participants will be classified as obese using the Asian population obesity threshold of BMI of ≥ 27.5.

Physical activity will be measured using CHNS physical activity questionnaire variables, including minutes per week spent walking or bicycling for transportation, occupational activity level, and time spent in sedentary behaviors. The CHNS also include variables that measure time spent walking to work or school, bicycling for commuting, and total transportation related activity time, which captures activity transportation patterns.

Urbanization exposure will be measured using community urbanization index scores, a composite indicator used in CHNS that includes measures of infrastructure development, population density, transportation systems, and economic activity. Additional indicators such as household ownership of motor vehicles or bicycles will also be used to measure transportation transitions associated with urbanization.

Key Outcomes and Variables

The primary outcome variable will be obesity status based on BMI classification. Key explanatory variables will include:

- Transportation related physical activity
- Community urbanization index score
- Household vehicle ownership
- Urban versus rural residence

Control variables will include age, sex, educational level, and household income, since these factors are associated with both physical activity behaviors and obesity risk.

Data Collection Personnel and Setting

All data used in this study were originally collected through the China Health and Nutrition Survey by trained interviewers and health technicians. Anthropometric measurements were taken during household visits and community survey locations using standardized

measurement protocols and equipment. Behavioral and demographic information was collected through questionnaires administered during in person interviews and adult participants

CHNS uses a multistage cluster sampling design across several provinces in China, which captures variation in socioeconomic development, infrastructure, and urbanization.

Source of Bias and Data Quality Concerns

One potential limitation is self report bias in physical activity measures, since participants may inaccurately report time spent walking or cycling for transportation.

Another concern is residual confounding, as obesity risk is also influenced by factors such as diet, food availability, and environmental conditions that may not be fully captured in the dataset.

Finally because the analysis relies on observational secondary data, the results can identify associations between urbanization related lifestyle factors and obesity prevalence but cannot establish causal relationships.

4. Measurement Instrument - Appendix

The primary measurement instrument is the China Health and Nutrition Survey (CHNS) questionnaire and anthropometric measurement protocol. The CHNS collects standardized information on demographic characteristics, physical activity behaviors, transportation habits and health outcomes. Below are examples of key variables and survey questions used in the study.

Anthropometric Measurements

Height and weight are measured by trained health technicians and used standardized equipment which includes height in centimeters measured with a portable stadiometer and

weight in kilograms measured with a calibrated digital scale. These values are used in calculation of Body Mass Index (BMI) with the standard formula:

$$BMI = weight\ (kg)/height(m^2)$$

Participants will be classified as obese using the Asian population obesity threshold of BMI of ≥ 27.5.

Physical activity and Transportation Access

The participants will report time spent in physical activity that doubles as a form of transportation. Also report vehicle ownership of households. Examples of questions:

1. How many minutes per day do you typically spend walking to work, school or others?
2. How many minutes per day do you spend bicycling for transportation?
3. How many days per week do you typically walk or bicycle for transportation?
4. Does your household own a motor vehicle?

5. Ethics and Feasibility Brief

This study relies on secondary data from the China Health and Nutrition Survey (CHNS), which presents minimal risk to participants because no new data are collected and there is no direct interaction with human subjects. The individuals included in the CHNS dataset had already provided informed consent at the time of the original data collection. As a result, the study does not introduce additional risk beyond what was already accounted for in the initial research process.

The primary ethical concern in this study is maintaining the privacy and confidentiality of participant information. Researchers must ensure that all data are de-identified and stored securely in password-protected systems. Additionally, access to the dataset must follow the usage agreements established by CHNS to prevent misuse of sensitive information. Because the study

involves analysis of existing, de-identified data and does not involve interaction with participants, it would likely qualify for exemption from Institutional Review Board (IRB) review.

In terms of feasibility, this study is practical because it utilizes an existing large-scale national dataset. However, there are some constraints, including obtaining access to the dataset, managing large and complex data files, and ensuring consistency of variables across different sources. Despite these challenges, secondary data analysis remains a feasible and efficient approach for examining population-level trends in obesity and urbanization.

6. Reflection on Methodological Trade-offs

One of the most challenging decisions in this study was selecting a study design that is most appropriate. While randomized controlled trials are considered the most ideal method for establishing causal relationships, they are not feasible for large study on structural changes such as urbanizations. Urban development and economic growth occur naturally at the societal level and can't be randomly assigned to individuals. Due to these disadvantages, an observational cross-sectional design using secondary data will be the most fit for this study. This will allow researchers to analyze large populations and analyze patterns between obesity prevalence and urbanization lifestyle changes. However, there are trade-offs as cross-sectional studies cannot establish causality and can be easily affected by confounders. Some condoning factors can be diet, environmental changes, etc. If given more time and funding, a longitudinal cohort study would strengthen the analysis of this study. As researchers would follow individuals over time to better evaluate how changes in physical activity and transportation behaviors influence obesity risk. This study demonstrated that real-world public health research often requires balancing scientific trials with limitations like feasibility and ethical considerations. Public health

researchers must make thoughtful decisions that produce meaningful evidence while also acknowledging the purpose of the study.

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